

KAVYA REDDY MARAKALA

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Portfolio : <https://kavya-portfolio-git-main-kavya-reddys-projects.vercel.app/>

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PROFESSIONAL SUMMARY

Backend Developer with 3+ years of experience in designing and implementing scalable backend systems across telecom and banking sectors. Possess expertise in using FastAPI, Flask, Django, and integrating with cloud services like AWS and GCP. Spearheaded the migration of over 100 Apache NiFi flows, achieving zero data loss, and reduced onboarding time by 70% through automation workflows. Engineered a high-performance chatbot with enhanced NLP capabilities, handling concurrent users effectively. Consistently deliver efficient, innovative solutions that drive organizational success and improve technical operations.

TECHNICAL SKILLS

- **Programming / Scripting Languages:** Python, Java, JavaScript, Golang, C++, Bash
 - **Frameworks & Libraries:** Django, Flask, FastAPI, Celery, RabbitMQ, Kafka, Dash, Plotly, React.js, Node.js
 - **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Cassandra
 - **AI & ML Tools:** OpenAI APIs, Llama (Meta), Hugging Face Transformers, RAG Pipelines, Vector Embeddings, PDF & Document Loaders, scikit-learn, TensorFlow
 - **API & Architecture:** RESTful APIs, GraphQL, Event-Driven Architecture, WebSockets
 - **Security & Authentication:** OAuth 2.0, JWT, Role-Based Access Control (RBAC), CyberArk
 - **DevOps & CI/CD:** Jenkins, GitHub Actions, GitLab CI, Docker, Kubernetes, Terraform, ELK Stack, AWS, GCP, Azure, AWS Lambda, Azure Functions, Ansible, Azure DevOps
 - **Data Processing & Machine Learning:** Apache Kafka, Apache Ranger, TensorFlow, scikit-learn, OpenAI APIs
 - **Testing & Monitoring:** Pytest, Selenium, Prometheus, Grafana, Azure Monitor
 - **Project Management & Agile Tools:** JIRA, Asana, Kanban, Scrum
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PROFESSIONAL EXPERIENCE

Verizon- Backend Developer

Basking Ridge, NJ

Feb 2024 – Present

- Built secure, low-latency APIs using FastAPI, Flask, and Django across Hadoop and GCP, improving data delivery speed by 30%.

- Implemented JWT-based authentication and role-based authorization for internal users, enhancing security coverage by 100%.
- Developed responsive UIs in React.js using reusable components, MUI, and Bootstrap, boosting development speed by 25%.
- Managed state with React hooks and optimized frontend load time with lazy loading, reducing page load times by 40%.
- Created dynamic, validated forms with React Hook Form, improving form submission accuracy by 35%.
- Integrated APIs using async/await in JavaScript for smoother user interactions.
- Migrated 100+ Apache NiFi flows with 0% data loss, improving reliability and reducing manual fixes by 50%.
- Built CI/CD pipelines for NiFi using Jenkins and Python, reducing deployment time by 60% and adopted by 50+ teams.
- Automated user access workflows across Hadoop, GCP, and Looker, cutting onboarding time by 70% and increasing team productivity.
- Built CLI tools with Cobra and Viper to streamline development tasks, adopted by 3+ internal teams.
- Engineered Kafka streaming pipelines with Sarama for real-time data ingestion and processing.
- Implemented event-driven architecture using Kafka and Pub/Sub to decouple services and improve fault tolerance.
- Led test automation using Pytest under BDD/TDD practices, achieving 85% test coverage.
- Architected microservices using Clean Architecture and domain-driven design (DDD) principles.
- Tuned system performance through profiling, caching, and optimization of high-traffic services.
- Collaborated with product and compliance teams to align features with security and financial regulations.
- Containerized services with Docker and orchestrated deployments via Kubernetes (AWS EKS).
- Automated infrastructure using Terraform and CI/CD via Jenkins and GitHub Actions.
- Monitored production systems with Prometheus, Grafana, and CloudWatch to maintain 99.9% uptime.

Elloe AI - Backend Developer Intern

Overland Park, KS

Aug 2023 – Dec 2023

- Developed a high-performance chatbot using Flask and FastAPI with NLP enhancements using TF-IDF, NLTK, and spaCy.
- Implemented asynchronous communication using Python's asyncio and multiprocessing, reducing response time by 35% and improving user concurrency handling.

- Integrated cloud platforms (AWS/GCP) for auto-scaled deployment via Kubernetes and Docker, increasing deployment efficiency by 40%.
- Wrote detailed unit and integration tests using Pytest achieving 95% test coverage, significantly enhancing code reliability.
- Spearheaded migration of 100+ Apache NiFi flows to target environments with zero data loss, enhancing data integrity and operational efficiency.
- Contributed to the design of event-driven workflows that supported async notification and retries.

Tata Consultancy Services - Python Developer

Chennai, India

Jul 2021 – July 2022

- Automated reporting and data transformation workflows using Python, AWS Glue, Lambda, and XML parsing.
- Developed internal APIs using Flask to support downstream systems, improving integration efficiency by 30%.
- Improved data consistency and reduced manual effort by 80% through scripted workflows, enhancing team productivity.
- Designed fault-tolerant data pipelines integrating Flexcube with AWS services (S3, RDS, CloudWatch), increasing data reliability by 50%.
- Built observability dashboards using Prometheus, Grafana, and integrated logging with ELK.
- Followed TDD methodology and participated in sprint ceremonies in an Agile environment.

PROJECTS (AI/ML)

Agentic Chatboard with LangGraph & LLMS

- Designed an end-to-end AI agent framework integrating LangGraph, LangChain, and OpenAI APIs for workflow orchestration.
- Built RAG pipelines with FAISS and ChromaDB for contextual search over PDFs and structured documents.
- Implemented document ingestion with Recursive Character Splitters & Embedding Models (OpenAI, Hugging Face, Llama) for efficient retrieval.
- Developed a Streamlit-based UI for interactive query handling and workflow visualization, enabling seamless debugging and model monitoring.
- Integrated multiple LLMs (OpenAI GPT, Llama, Hugging Face models) with real-time routing logic for optimal performance.

Intelligent Document Assistant with VectorDBs & RAG

- Built an AI-powered assistant to process large-scale PDFs, knowledge bases, and structured data using LangChain & Hugging Face embeddings.
 - an advanced retrieval-augmented generation (RAG) pipeline with ChromaDB and FAISS, improving accuracy of answers by 45%.
 - Implemented custom PDF & web loaders, recursive splitting strategies, and semantic chunking for better context retention.
 - Deployed as a containerized microservice on AWS Lambda + API Gateway, serving as a backend for enterprise search applications.
 - Added monitoring dashboards for LLM queries using LangSmith & Prometheus to track latency and token usage.
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EDUCATION

University of Central Missouri

GPA: 3.8 / 4.0

M.S. in Computer Science

Aug 2022 – Dec 2023

GNITS

GPA: 3.2/4

B.Tech in ECE

July 2017 - July 2021